

Greibach Normal form (GNF)

A CFG is said to be in GNF, if the production are in the form

$$A \rightarrow a\alpha$$

$$\text{i.e. } A \rightarrow aA_1A_2\ldots$$

$$A \rightarrow a.$$

where α can be one or more non-terminal.

Steps to Convert CFG to GNF.

- ① Simplify the CFG. i.e. Remove ϵ , unit, Reducible Grammar.
- ② Check whether the grammar is in CNF or Not.
If it is not in CNF, then Convert it into CNF.
- ③ Each Non-terminal are replace by $A_1, A_2, A_3, \dots$ in horizontal order of each production.
- ④ If the production is in the form of

$$A_i \rightarrow A_j \beta \text{ then}$$

check if $i < j \rightarrow$ leave the production now & ^{do} obtain substitution after getting A_j value ~~substitution~~.

if $i > j \rightarrow$ do the Substitution immediately.

if $i = j \rightarrow$ (left recursion) eliminate it.

Ex Convert CFG to ~~GNF~~ GNF.

$$S \rightarrow CA \mid BB$$

$$B \rightarrow b \mid SB$$

$$C \rightarrow b \mid$$

$$A \rightarrow a \mid$$

} in GNF

it is in simplified form.
as there is no any ϵ , unit production & no useless.
it is also in CNF.

Sol Replace S with A_1
 $C \rightarrow A_2$
 $A \rightarrow A_3$
 $B \rightarrow A_4$

$$A_1 \rightarrow A_2A_3 \mid A_4A_4$$

$$A_4 \rightarrow b \mid A_1A_4$$

$$A_2 \rightarrow b$$

$$A_3 \rightarrow a.$$

Consider $A_1 \rightarrow A_2 A_3 \mid A_4 A_5$

$i=1, j=2 \quad i < j$ leave it now,

we get value of A_2 then we replace

Consider $A_4 \rightarrow A_1 A_5 \mid b$

$i=4, j=1 \quad i > j$ then do substitution.

Substitute A_1 value in $A_4 A_5$ (i.e. $A_1 \rightarrow A_2 A_3 \mid A_4 A_5$)

$A_4 \rightarrow A_2 A_3 A_5 \mid A_4 A_5 A_5 \mid b$.

Here again $i=4, j=2$ do the substitution

Substitute value of A_2 in above A_5 (i.e. $A_2 \rightarrow b$)

$A_4 \rightarrow \underline{b A_3 A_5} \mid A_5 A_5 A_5 \mid b$.

$i=4, j=4$ so eliminate the left recursion.

eliminate
left
recursion.

$A \rightarrow A \alpha \mid \beta$

then $\begin{cases} A \rightarrow \beta A' \\ A' \rightarrow \alpha A' \mid \epsilon \end{cases}$

(eliminate ϵ)

Here

~~$A_4 \rightarrow A_4 A_5 A_5$~~

$A_4 \rightarrow b A_3 A_5 \mid A_5 A_5 A_5 \mid b$

Here

$A \rightarrow \beta$

$A \rightarrow \alpha$

~~$A_4 \rightarrow A_4 A_5 A_5 \mid b A_3 A_5 \mid b$~~

$A_4 \rightarrow \underline{A_4 A_5 A_5} \mid b \mid b A_3 A_5$
 $A \quad \alpha \quad B_2 \quad B_1$

$\begin{cases} Z_4 \rightarrow \alpha \mid \alpha Z_4 \\ A_4 \rightarrow \beta \mid \beta Z_4 \end{cases}$ formula.

Here Z_4 is A'

$\begin{cases} A_4 \rightarrow \underline{b A_3 A_5 Z_4} \mid \underline{b Z_4} \mid b A_3 A_5 \\ Z_4 \rightarrow \underline{A_4 A_5 Z_4} \mid \epsilon \end{cases}$ (Here eliminate ϵ)
 $Z_4 \rightarrow \epsilon$

So replace each Z_4 by ϵ .

$A_4 \rightarrow b A_3 A_5 \mid b \mid b A_3 A_5 \mid b$ $\rightarrow$ is in GNF form.

~~$A_4 \rightarrow b A_3 A_5 \mid b$~~

$Z_4 \rightarrow \underline{A_4 A_5 Z_4} \mid \underline{A_4 A_5}$

$\rightarrow$ Not in GNF so we convert it in to GNF.

Substitute A_4 in Z_4

$$Z_4 \rightarrow bA_3A_4Z_4 | A_4A_4$$

$$Z_4 \rightarrow bA_3A_4Z_4A_4Z_4 | bA_3A_4Z_4A_4 | bZ_4A_4Z_4 | bZ_4A_4$$

$$| bA_3A_4A_4Z_4 | bA_3A_4A_4 | \cancel{bA_3A_4Z_4} | bA_4$$

it is in GNF.

Consider

$$A_1 \rightarrow A_2A_3 | A_4A_4$$

$$\rightarrow bA_3 | bA_3A_4Z_4A_4 | bZ_4A_4 | bA_3A_4Z_4 | bA_4$$

$$A_2 \rightarrow b$$

$$A_3 \rightarrow a$$

$$A_4 \rightarrow bA_3A_4Z_4 | bZ_4 | bA_3A_4 | b$$

$$Z_4 \rightarrow \text{above given.}$$

Now it is in GNF.

$$A_4 = bA_3A_4 | \underbrace{A_4A_4A_4}_A | \underbrace{bA_3A_4}_{B_1} | \underbrace{b}_{B_2}$$

$$BB \rightarrow \alpha | \alpha B$$

$$A \rightarrow B | BB$$

$$Z_4 \rightarrow \alpha | \alpha Z_4$$

$$A \rightarrow B | BZ_4$$

$$A_4 = bA_3A_4 | b | bA_3A_4 | bZ_4$$

$$Z_4 \rightarrow A_4A_4 | A_4A_4Z_4$$

Same Result

Construct a grammar in GNF which is Keedy equivalent to the grammar.

$$S \rightarrow AA|a$$

$$A \rightarrow ss|b.$$

Sol

Step 1 - No elimination of null, useless, unit already in simplified form.

Step 2 Rename Variable

$S \leftarrow A$ & $A \leftarrow A_1$
 $A_1 \leftarrow A_2$ respectively.

So

$$A_1 \rightarrow \underline{A_2 A_2} | \underline{a}$$

$$\Rightarrow A_2 \rightarrow \underline{A_1 A_1} | \underline{b} \quad \checkmark \text{ in GNF.}$$

Step 3 - Every production of the form.

$A_i \rightarrow A_j \alpha$ with $i > j$ must be modified to make $i \leq j$.
 So A_2 must be modified.

$A_2 \rightarrow A_1 A_1$ should be modified.
 Use Rule 1 is

Let $A \rightarrow B\alpha$ be some production of A
 & let $B \rightarrow B_1 | B_2 | B_3 \dots | B_n$ then we can write
 $A \rightarrow B_1\alpha | B_2\alpha | B_3\alpha | \dots | B_n\alpha$.

So substitute for A_1

So

$$A_2 = A_2 A_2 A_1$$

$$A_2 = a A_1 | b$$

Now Rewrite all the production.

$$\begin{aligned} [A_1 \rightarrow A_2 A_2 | a] \\ A_2 \rightarrow A_2 A_2 A_1 | \underline{QA_1} | b \end{aligned} \quad \begin{array}{l} \text{in} \\ \text{GNF} \end{array}$$

Consider

$$A_2 \rightarrow \frac{A_2 A_2 A_1}{A} | \frac{QA_1}{R_1} | \frac{b}{R_2}$$

Left Recursion.

use Rule 2

$$\begin{aligned} \text{Let } A &\rightarrow A\alpha_1 | A\alpha_2 | A\alpha_3 | \dots | A\alpha_r \\ &B \rightarrow B_1 | B_2 | B_3 | \dots | B_r \end{aligned}$$

then introduce a new variable B as

Remember

$$\begin{aligned} B &\rightarrow \alpha_i | \alpha_i B \\ B &\rightarrow B_i | B_i B \end{aligned} ; 1 \leq i \leq r$$

$$A \rightarrow A\alpha | B$$

$$\begin{aligned} \text{in } R \\ A &\rightarrow A\alpha | B \\ A &\rightarrow BA' \\ A' &\rightarrow \alpha A' \end{aligned}$$

so using rule 2

$$\begin{aligned} B &\rightarrow A_2 A_1 | A_2 A_1 B \\ A &\rightarrow \underline{QA_1} | b | QA_1 B | bB \end{aligned}$$

This is A_2 which is in GNF

Now resulting production is

$$\begin{aligned} A_1 &\rightarrow A_2 A_2 | a \quad \leftarrow \text{put} \\ A_2 &\rightarrow \underline{QA_1} | b | QA_1 B | bB \quad \leftarrow \text{put} \\ B &\rightarrow A_2 A_1 | A_2 A_1 B \quad \leftarrow \text{put} \end{aligned}$$

~~For abate A122~~

A's & B's production can be converted to GNF with the help of A2's production using Rule 1

$A_1 \rightarrow$

$A_2 \rightarrow \cancel{QA_1b} | \cancel{b} | \cancel{QA_1B} | \cancel{bB}$

$A_1 \rightarrow QA_1A_2 | \cancel{bA_2} | \cancel{QA_1BA_2} | \cancel{bBA_2} | \underline{a}$

$A_2 \rightarrow QA_1 | \underline{b} | \underline{QA_1B} | \underline{bB}$

$B \rightarrow B \cdot QA_1A_1 | bA_1 | QA_1BA_1 | bBA_1 |$

$QA_1A_1B | bA_1B | \underline{QA_1BA_1B} | \underline{bBA_1B}$

where $V = (A_1, A_2, B)$

$T \rightarrow (a, b)$

$S \rightarrow A_1$

$P = qrc$

$A_1 \rightarrow$

$A_2 \rightarrow$

$A_3 \rightarrow$

This is GNF.

OR $A_2 \rightarrow \frac{A_2A_2A_1}{A} | \frac{QA_1}{B_1} | \frac{b}{B_2}$

$A_2 \rightarrow \underline{QA_1B} | \underline{bB}$

$B \rightarrow A_2A_1B$

new $A_1 \rightarrow \underline{A_2A_2} | \underline{a}$ — privious

$B \rightarrow \underline{QA_1BA_1B} | \underline{bBA_1B}$

$A_1 \rightarrow \underline{QA_1BA_2} | \underline{bBA_2} | \underline{a}$

$A \rightarrow \cancel{AA} | B$

$A \rightarrow BB$

$B \rightarrow \alpha B$

Convert the CFG into GNF.

$$\begin{aligned} S &\rightarrow AB \\ A &\rightarrow BS|b \\ B &\rightarrow SA|a \end{aligned}$$

First step -

Any grammar is already in Chomsky Normal form if not convert into CNF.

(i) Second step - give Non-terminals separate naming.

$$\begin{aligned} S &\rightarrow A_1 \\ A &\rightarrow A_2 \\ B &\rightarrow A_3 \end{aligned}$$

So our grammar become.

$$\begin{aligned} A_1 &\rightarrow A_2 A_3 \checkmark \\ A_2 &\rightarrow A_3 A_1 | b \checkmark \\ A_3 &\rightarrow A_1 A_2 | a \times \end{aligned} \quad \begin{array}{l} \swarrow \\ \rightarrow A_i < A_j \\ \searrow \end{array}$$

for $A_3 \rightarrow A_1 A_2 | a$.

$A_3 > A_1$ so substitute & change the production to the form $A_i \rightarrow A_j \quad i < j$.

Put the value of A_1 & A_2 in A_3 .

We get $A_3 \rightarrow A_2 A_3 A_2 | a$.

$$\underline{A_3} \rightarrow \underline{A_3} A_1 A_3 A_2 | a | b A_3 A_2$$

Step 3

Here there is recursion $A_3 = A_3$

i.e. $A_i = A_j$
So eliminate left recursion.

$$A_3 \rightarrow \underbrace{A_3 A_1 A_3 A_2}_A | \underbrace{a}_B | \underbrace{b A_3 A_2}_B$$

Take another variable Z_3 .

$$A_3 \rightarrow A Z_3 | b A_3 A_2 Z_3$$

$$Z_3 \rightarrow A_1 A_3 A_2 Z_3 | \epsilon$$

Now eliminate epsilon production.

$$A_3 \rightarrow A Z_3 | b A_3 A_2 Z_3 | a | b A_3 A_2 \leftarrow \text{it is in GNF}$$

$$Z_3 \rightarrow A_1 A_3 A_2 Z_3 | A_1 A_3 A_2$$

$A \rightarrow A\alpha \beta$
$A' \rightarrow \beta A'$
$A' \rightarrow \alpha A'$

Now Consider

$$A_2 \rightarrow A_3 A_1 | b$$

Put value of A_3 in above we get

$$A_2 \rightarrow a_3 A_1 | b A_3 A_2 Z_3 A_1 | a A_1 | b A_3 A_2 A_1 | b \rightarrow \text{in GNF}$$

Now Consider

$$A_1 = A_2 A_3$$

Put value of A_3 & A_2 in above

$$A_1 \rightarrow b A_3 | a_3 A_1 A_3 | b A_3 A_2 Z_3 A_1 A_3 | a A_1 A_3 | b A_3 A_2 A_3 \rightarrow \text{in GNF}$$

Now Consider

$$Z_3 \rightarrow A_1 A_3 A_2 Z_3 | A_1 A_3 A_2$$

Put the value of A_1 Here

$$\begin{aligned} Z_3 \rightarrow & b A_3 A_3 A_2 Z_3 | a_3 A_1 A_3 A_3 A_2 Z_3 | b A_3 A_2 Z_3 A_1 A_3 A_3 A_2 Z_3 | \\ & a A_1 A_3 A_3 A_2 Z_3 | b A_3 A_2 A_1 A_3 A_3 A_2 Z_3 | \\ & | b A_3 A_3 A_2 | a Z_3 A_1 A_3 A_3 A_2 | b A_3 A_2 Z_3 A_1 A_3 A_3 A_2 \\ & | a A_3 A_3 A_2 | b A_3 A_2 A_1 A_3 A_3 A_2 \rightarrow \text{in GNF} \end{aligned}$$

✓ Convert the following CFG into GNF.

$$S \rightarrow AO$$

$$A \rightarrow OB$$

$$B \rightarrow AO \mid 1$$

Soln

Above grammar having No any ϵ or unit production.
But it is No in Chomsky Normal form
So convert it into CNF.

Replace O with X

So our grammar become.

$$S \rightarrow AX$$

$$A \rightarrow XB$$

$$B \rightarrow AX \mid 1$$

$$X \rightarrow O$$

Now it is in CNF.

Now Replace S with A_1
A with A_2
X with A_3
B with A_4 .

We get

$$A_1 \rightarrow A_2 A_3 \checkmark$$

$$A_2 \rightarrow A_3 A_4 \checkmark$$

$$A_4 \rightarrow A_2 A_3 \mid 1$$

$$A_3 \rightarrow O$$

$$\text{Take } A_4 \rightarrow A_2 A_3 \mid 1$$

Put value of $A_2 = A_3 A_4$ in above.

$$A_4 \rightarrow A_3 A_3 \mid 1$$

$$\text{Put } A_3 \rightarrow O$$

$$A_4 \rightarrow O A_3 \mid 1$$

$$\text{Now } A_1 \rightarrow A_2 A_3$$

$$\text{Put } A_2 \rightarrow A_3 A_4$$

$$A_1 \rightarrow A_3 A_4 A_3$$

$$\text{Put } A_3 \rightarrow O$$

$$A_1 \rightarrow O A_4 A_3$$

$$\text{Now } A_2 \rightarrow A_3 A_4$$

$$A_2 \rightarrow O A_4$$

So final GNF

$$A_1 \rightarrow O A_4 A_3$$

$$A_2 \rightarrow O A_4$$

$$A_4 \rightarrow O A_3 \mid 1$$

$$A_3 \rightarrow O$$

Convert the following CFG into GNF.

$$S \rightarrow A|AB$$

$$A \rightarrow aA|e$$

$$B \rightarrow bB|b$$

Soln

First Remove ϵ -production

$A \rightarrow \epsilon$ Nullifier.

$$w_0(A) \rightarrow \{S, A\}$$

$$w_1(A) \rightarrow$$

$$w_1 = \{A\} \text{ direct}$$

$$w_2 \rightarrow \{S, A\} \text{ indirect}$$

$$w_3 \rightarrow w_3 \text{ skip}$$

So Expand production.

$$S \rightarrow A|AB|B|e$$

$$A \rightarrow aA|a$$

So updated Grammar will be.

$$S \rightarrow A|AB|B|e$$

$$A \rightarrow aA|a$$

$$B \rightarrow bB|b$$

Now Eliminate unit production.

$$S \rightarrow a|aA$$

$$S \rightarrow aA|a|AB|bB|b|e$$

$$A \rightarrow aA|a$$

$$B \rightarrow bB|b$$

$$w_0(S) \rightarrow \{S, A, B\}$$

$$w_1(S) \rightarrow w_0(S) \cup \phi$$

$$= w_0(S)$$

Now Convert it into Chomsky Normal form.

Replace a with x

b with y

$$S \rightarrow xA|a|AB|yB|b|e$$

$$A \rightarrow xA|a$$

$$B \rightarrow yB|b$$

$$x \rightarrow a$$

$$y \rightarrow b$$

Now Replace

$$S \rightarrow A_1$$

$$x \rightarrow A_2$$

$$A \rightarrow A_3$$

$$B \rightarrow A_4$$

$$y \rightarrow A_5$$

$$A_1 \rightarrow A_2 A_3 | a | A_3 A_4 | A_5 A_4 | b | e$$

$$A_3 \rightarrow A_2 A_3 | a$$

$$A_4 \rightarrow A_3 A_4 | b$$

$$A_2 \rightarrow a \quad A_5 \rightarrow b$$

$$\begin{aligned} A_1 &\rightarrow A_2 A_3 \mid A_3 A_4 \mid A_5 A_4 \mid a \mid b \mid \epsilon \quad (\text{Leave it}) \\ A_2 &\rightarrow A_2 A_3 \mid a \quad (\text{No Replacement}) \\ A_4 &\rightarrow A_5 A_4 \mid b \quad (\text{Leave it}) \end{aligned}$$

Consider $A_3 \rightarrow A_2 A_3 \mid a$.

As $i > j$ No replacement.
 $3 > 2$.

Put the Put $A_2 \rightarrow a$.

$$A_3 \rightarrow a A_3 \mid a$$

$$A_1 \rightarrow a A_3$$

$$A_1 \rightarrow A_3 A_4$$

$$A_1 \rightarrow a A_3 A_4 \mid a A_4$$

$$A_1 \rightarrow b A_4$$

$$\left\{ \begin{aligned} A_1 &\rightarrow a A_3 \mid a A_3 A_4 \mid a A_4 \mid \\ &\quad b A_4 \mid a \mid b \mid \epsilon. \end{aligned} \right.$$

$$A_4 \rightarrow A_5 A_4 \mid b$$

$$A_4 \rightarrow b A_4 \mid b$$

So final GNF will be.

$$A_1 \rightarrow a A_3 \mid a A_3 A_4 \mid a A_4 \mid b A_4 \mid a \mid b \mid \epsilon$$

$$A_2 \rightarrow a$$

$$A_3 \rightarrow a A_3 \mid a$$

$$A_4 \rightarrow b A_4 \mid b$$

$$A_5 \rightarrow b$$